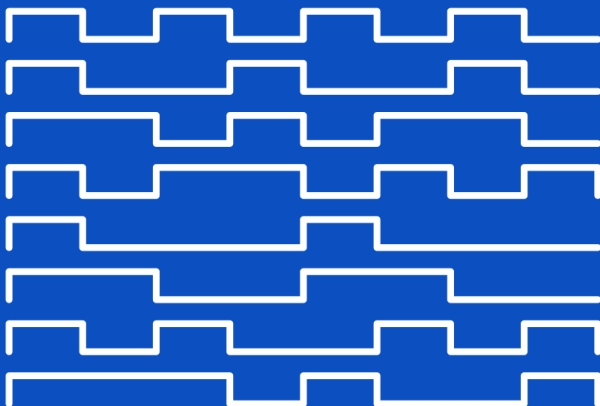




CLOCK

User Manual

Eight-channel clock & gate generator · 10 HP

Independent · One Clock · Divider Bank

Design & firmware Axel Napolitano
Beta · firmware 0.19.0-beta.14
Manual CC BY-NC 4.0

From Munich with



1 Start here

CLOCK is an eight-channel Eurorack clock and gate-rhythm generator. It can be a conventional master clock, build eight independent rhythmic streams, duplicate one shared clock to all outputs, or act as an eight-output divider bank.

The three ways **CLOCK** can route its outputs

Topology	What the eight outputs do
Independent	Each channel independently uses Off, Clock, Euclid or Sequencer.
One Clock	One shared clock configuration drives all eight outputs. This is the factory topology.
Divider Bank	One shared clock feeds eight fixed divider outputs from a selectable family.

V1 FEATURE FREEZE — FIRMWARE 0.19.0-BETA.14

This is prerelease hardware and firmware. The software timing model, presets and display transports are implemented, but external SYNC/RST hardware routing and physical timing sign-off are not final. Treat INTERNAL clocking as the verified operating path until HIL validation is complete.

One rule that explains the design

All eight outputs share one deterministic master timeline. PLAY/PAUSE, STOP, Tap Tempo and global phase are master-level operations. Independent channels change what they derive from that timeline; they do not run eight unrelated transports.

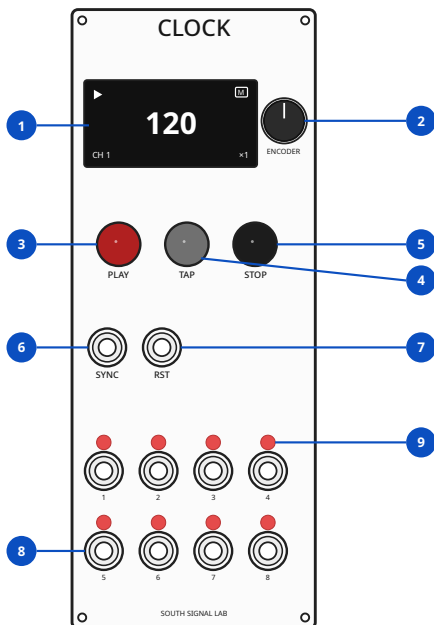
Why no general CV modulation?

CLOCK deliberately stays in the digital timing, gate and trigger domain. Hardware Rev 1 has dedicated SYNC and RST inputs, but no general parameter-CV inputs, analog modulation outputs or CV modulation matrix. This is a product boundary, not a missing V1 feature.

A quality eight-channel analog modulation-output path would require an eight-channel 16-bit DAC-class solution plus two quad output-op-amp stages. A 12-bit MCP-class shortcut is not considered adequate for this product direction. The current project estimate is roughly EUR 30-40 additional BOM cost before the added fine-pitch SMD assembly, PCB, calibration and validation burden. CLOCK therefore spends its complexity budget on precise digital timing and rhythm; later cross-channel interaction can remain internal and event-based.

2 Panel, power and boot

CLOCK uses one 128×64 OLED, one push encoder, three transport/navigation buttons, SYNC and RST inputs, and eight gate outputs with activity LEDs.



Panel element	Purpose
1 - OLED	Performance view, menus and editors
2 - Encoder	Turn to select/change; short and long push have different jobs
3 - PLAY	Start or pause the master transport
4 - TAP	Tap Tempo; also a modifier for Settings and mode selection
5 - STOP / BACK	STOP from Performance; BACK/cancel elsewhere
6 - SYNC IN	External clock input (prerelease hardware path)
7 - RST IN	External reset/phase input (prerelease hardware path)
8 - OUT 1–8	Gate/clock outputs; LED mirrors each logical output
9 - Activity LEDs	One 3 mm indicator above each output; shows output activity

Installation

Use a standard Eurorack 2×8 power cable. Align the red stripe with –12 V and verify the connector orientation before applying power.

Safe boot

Every power-up enters STOP. The gate buffer remains disabled while the firmware initializes; channel source pins are driven LOW before the output stage is enabled. A previously saved PLAY state is never resumed automatically.



The boot screen lasts about one second and shows CLOCK, the firmware version and a two-pixel progress bar. Holding the encoder push continuously through the boot screen launches the selected Easter egg; see section 19.

3 Two-minute quick start

FACTORY STATE

One Clock · 120 BPM · 4/4 · INTERNAL source · STOP · ×1 rate · 0% Swing · 10 ms gates · 0% Phase · Humanize OFF · Clock screensaver after 2 min in STOP, dim after 5 min, OLED off after 10 min.

Get eight identical clocks

- 1. Power the rack. CLOCK must finish booting in STOP.
- 2. Patch any of OUT 1–8 to modules that accept a clock or trigger.
- 3. Turn the encoder on Performance to set the master BPM.
- 4. Press PLAY. In the factory One Clock topology, all eight outputs now follow the same clock.
- 5. Press PLAY again to pause. Press STOP / BACK from Performance to stop and reset global phase.



Move from One Clock to independent rhythms

- 1. Short-press the encoder to open the global overview.
- 2. Hold TAP and turn the encoder. The six-function palette appears.
- 3. Choose CLOCK, EUCLID or SEQUENCER. Releasing TAP opens CHANGE MODE? with NO selected.
- 4. Select YES and press the encoder. Choosing one of those channel functions switches to Independent operation for the selected channel context.
- 5. From Performance, short-press the encoder to select among channels 1–8.

SAFE CONFIRMATION

Changing between channel functions and global output topologies is never applied merely by brushing past a choice. CHANGE MODE? defaults to NO.

4 The control language

Most of CLOCK becomes predictable once the distinction between a short encoder press, a long encoder press and a TAP chord is clear.

Gesture	What it means
Encoder turn	Performance: change master BPM. Overview: move highlight. Settings/editor: move cursor or change the edited value.
Encoder short press	Performance: open overview. Independent overview: confirm highlighted channel and return. Settings/editor: enter, confirm or toggle a sequencer step.
Encoder long press (~650 ms)	From Performance or overview: open the settings belonging to the current/highlighted channel or global topology.
TAP + encoder press	From Performance: open the root Settings tree.
Hold TAP + encoder turn	From an overview: open and move through the six-function mode palette. Release TAP to commit the highlighted choice.
Button	Contextual behavior
PLAY	Performance: play/pause. Sequencer editor: next 16-step page.
TAP	Performance: Tap Tempo. Overview: modifier for mode selection. Sequencer editor: previous 16-step page.
STOP / BACK	Performance: STOP + global phase reset. Elsewhere: back/cancel.

Editing a value

- 1. Turn to select a row.
- 2. Press to edit or open it.
- 3. Turn to change the value.
- 4. Press to confirm.

Only the value being edited is inverted. BACK leaves the current page or cancels the current confirmation workflow.

SHORT OR LONG?

A short encoder push from Performance is for orientation: it opens the overview. A long push is the direct route to the settings that belong to the current context.

5 Read the Performance screen

Performance is intentionally sparse. It shows the state needed while patching; detailed editing lives on dedicated pages.



Header

- **M / S:** filled M = master; outlined S = slave.
- **Padlock:** shown beside S when external timing is locked.
- **CHn + pictogram:** selected Independent channel and its generator.
- **Meter:** master meter in the center.
- **Transport:** PLAY, PAUSE or STOP at the right.

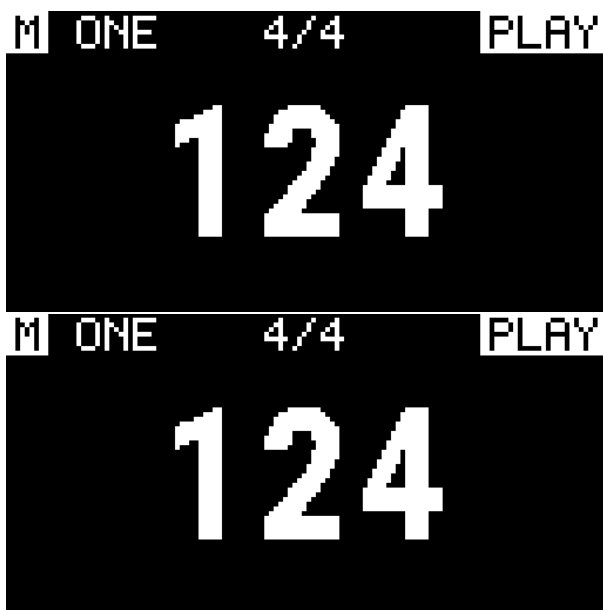
Tempo area

The BPM number stays mathematically centered. A non-zero Swing value appears at the left; a Clock rate other than $\times 1$ appears at the right. Zero Swing and $\times 1$ are omitted to keep the display quiet.

Pattern feedback

Euclid and Sequencer use the lower display for live pattern feedback. Filled cells are gates, outlined cells are rests, and an underline marks the current step. A sequence longer

than 16 steps adds a block indicator for the active 16-step segment.



6 Select channels and change functions

Independent channel overview

In Independent topology, short-press the encoder from Performance to open SELECT CHANNEL. The 2×4 grid shows channels 1–8 and their pictograms. Turn to move the inverted highlight. Short-press to select that channel and return to Performance; long-press to open the highlighted channel settings.



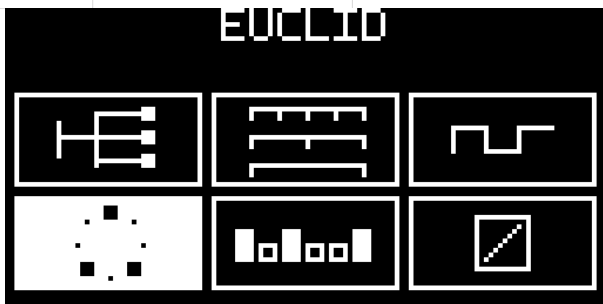
Global overview

One Clock and Divider Bank do not pretend to have eight independent channel menus. Their overview shows the active global topology. A short press returns to Performance; a long press opens that topology's settings.

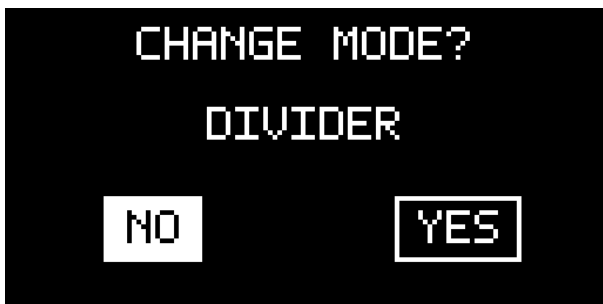
Six-function palette

Function	Scope	Destination after confirmation
Off	Selected channel / Independent	Performance (---)
Clock	Selected channel / Independent	Performance
Euclid	Selected channel / Independent	Euclid settings

Function	Scope	Destination after confirmation
Sequencer	Selected channel / Independent	16-step editor
One Clock	Global topology	One Clock settings
Divider Bank	Global topology	Divider family setting



Hold TAP and turn in the overview, then release TAP. If the highlighted function differs from the current one, CHANGE MODE? appears with NO selected by default.



7 Transport and tempo

PLAY, PAUSE and STOP

Action	Result
PLAY from STOP or PAUSE	Transport enters PLAY.
PLAY while playing	Transport enters PAUSE and preserves position.
STOP / BACK from Performance	Transport enters STOP and resets global phase according to reset policy.

Set BPM with the encoder

Turn the encoder on Performance. Manual tempo changes obey the user MIN BPM and MAX BPM limits. Factory limits are 20...999 BPM; those limits can themselves be set anywhere within the technical 1...999 BPM range.

Tap Tempo

Tap TAP repeatedly on Performance. The first tap starts a measurement without visual feedback; the second produces the first estimate and starts a four-frame 8×8 shrinking-dot animation immediately to the right of the BPM numerals. Every following tap in the same sequence restarts that animation while the estimator continues to use a rolling average of recent intervals. A valid Tap Tempo result selects INTERNAL as the clock source. The sequence remains valid for one beat at the configured MIN BPM; after a longer pause the next tap starts a new measurement and is again visually silent.

WHY TAP SELECTS INTERNAL

Tap Tempo is an explicit request to make CLOCK the timing authority. External synchronization is therefore left rather than silently competing with the tapped tempo.

8 Independent channels

Independent topology lets each of the eight outputs choose Off, Clock, Euclid or Sequencer while remaining phase-related to the same master timeline.

What every active channel can set

Parameter	Range / meaning
RATE	$\div 32, \div 24, \div 16, \div 12, \div 8, \div 6, \div 4, \div 3, \div 2, \times 1, \times 2, \times 3, \times 4, \times 6, \times 8, \times 12, \times 16, \times 24, \times 32$
Numerator / denominator	1...16 each; enables rational ratios such as 2:3 or 3:2
SWING	0...50%
PROBABILITY	0...100%
GATE	1, 2, 5, 10, 20, 50 or 100 ms
PHASE	0...99%
RESET	GLOBAL or FREE
MUTE	Suppress output without discarding the generator setup

OFF versus MUTE

OFF	MUTE
Removes the channel from event scheduling.	Keeps the generator running but suppresses its output.
Forces logical output LOW.	Useful for temporary silence.
Stored configuration remains available for later reuse.	Mode and settings remain active.

RESET POLICY

GLOBAL channels follow global phase resets and external alignment. FREE channels keep their local cycle position when global phase is reset.

9 Clock mode

Clock mode emits a gate at every eligible channel event. Use it for conventional clock trees, multiplied clocks, polymetric relationships and phase-shifted timing.

Clock-specific settings

Setting	Range / use
Meter beats	1...16
Meter unit	2, 4, 8 or 16
RATE DIV/MULT	Conventional integer divisions and multiplications
Ratio	Numerator/denominator for exact rational relationships

The timing engine retains fractional remainders instead of truncating them, so rational ratios do not accumulate drift simply because an interval is not an integer number of scheduler quanta.

Example: 3 against 2

- 1. Select a Clock channel.
- 2. Open RATE.
- 3. Set numerator to 3 and denominator to 2.
- 4. Keep RESET = GLOBAL if it must return to the common downbeat.

10 Euclid mode

Euclid distributes a chosen number of hits as evenly as possible over a cycle. It is useful for percussion, accents and repeating rhythmic structures that remain easy to reshape.

Parameter	Range
STEPS	1...64
HITS	0...STEPS
ROTATE	0...STEPS – 1

At RATE $\times 1$, each Euclid step is a sixteenth note. In 4/4, 16 steps therefore span one bar. RATE scales that step grid. Reducing STEPS automatically clamps HITS and, where needed, rotation to valid values.



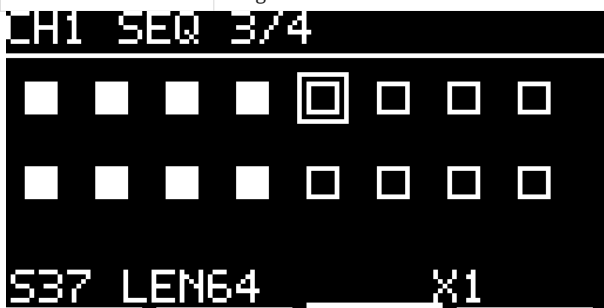
Build a 16-step pattern

- 1. Choose Euclid for a channel.
- 2. Set STEPS = 16.
- 3. Set HITS to the desired density.
- 4. Use ROTATE to move the pattern around the bar without changing the number of hits.
- 5. Use PROBABILITY for controlled variation without rewriting the Euclid pattern.

11 Sequencer mode

Each Sequencer channel stores a 64-bit gate pattern and an active length from 1 to 64 steps. At RATE $\times 1$, one step is a sixteenth note.

Page item	What it does
EDITOR	Open the step editor
LENGTH	1...64 active steps
ROTATE	Rotate inside the active length
INVERT	Invert every step inside the active length
CLEAR	Remove all gates
FILL ALTERNATE	Fill alternating steps, beginning with step 1
COPY	Copy the 64-bit pattern
PASTE	Paste and clamp to the destination length



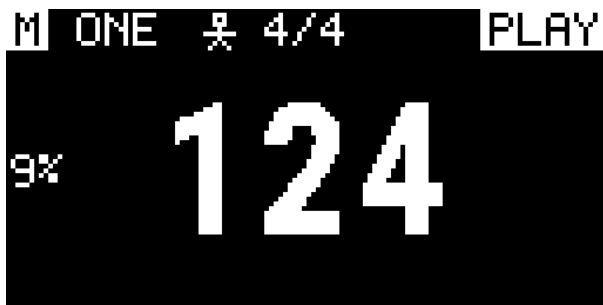
Editor controls

- **Encoder turn** — move the cursor through active steps.
- **Encoder press** — toggle the selected gate.
- **PLAY** — next 16-step page.
- **TAP** — previous 16-step page.
- **STOP / BACK** — return to Sequencer settings.

The four editor pages are 1–16, 17–32, 33–48 and 49–64. The cursor cannot move beyond the active sequence length.

12 One Clock

One Clock is the simplest global topology: a single clock configuration drives all eight physical outputs. This is the factory state.



Setting	Range
DIV/MULT	÷32 ... ×32
Numerator / denominator	1...16 each
Swing	0...50%
Gate length	1, 2, 5, 10, 20, 50 or 100 ms
Phase	0...99%
Humanize	OFF, 250, 500, 1000 or 2000 µs

Humanize

Humanize adds a small deterministic per-output timing displacement while preserving the common restart/downbeat. It exists only in One Clock; OFF gives exact coincidence across the eight outputs.

USE ONE CLOCK WHEN

You need one authoritative clock on several destinations and do not want eight separate channel configurations.

13 Divider Bank

Divider Bank derives eight deterministic clocks from the master timeline. Only the divider family and shared gate length are editable.

M

DIU

4/4

PLAY

124

×1

÷2

÷3

÷5

÷7

÷11

÷13

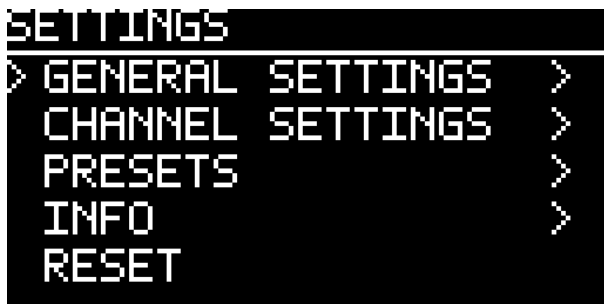
÷17

Family	OUT 1...8
POW2	×1 · ÷2 · ÷4 · ÷8 · ÷16 · ÷32 · ÷64 · ÷128
INT	×1 · ÷2 · ÷3 · ÷4 · ÷5 · ÷6 · ÷7 · ÷8
PRIME	×1 · ÷2 · ÷3 · ÷5 · ÷7 · ÷11 · ÷13 · ÷17

The Performance screen shows all eight assignments, so there is no synthetic per-channel settings page in this topology.

14 Settings map

Hold TAP and press the encoder from Performance to open the root Settings tree. The hierarchy is intentionally shallow and context-sensitive.



Root item	Contains / does
GENERAL SETTINGS	CLOCK, SYNC, SCREENSAVER, ENCODER DIR, ORIENTATION
CHANNEL SETTINGS	Current Independent channel, One Clock or Divider Bank settings
PRESETS	CURRENT auto-save, Load Preset, Save Preset, Templates
INFO	Name, firmware version, author, licenses, update QR
RESET	Apply global phase reset; this is not Factory Reset

GENERAL SETTINGS → CLOCK

Setting	Range
BPM	Current master tempo
MIN BPM	1...MAX; factory 20
MAX BPM	MIN...999; factory 999
Meter beats	1...16
Meter unit	2, 4, 8 or 16

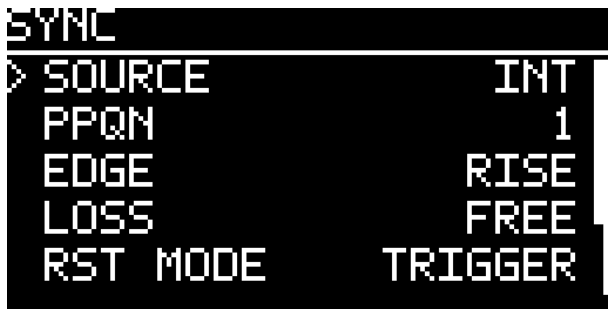
Changing MIN or MAX clamps the current manual tempo if necessary. External synchronization deliberately follows

the external timing source rather than the manual BPM limit.

ENCODER DIR selects NORMAL or REVERSED user-facing rotation after detent decoding. ORIENTATION selects 0 DEG or 180 DEG. At 180 DEG the firmware rotates the complete 128×64 transfer framebuffer before sending it to the OLED while the controller remains in the proven A1/C8 scan orientation. This avoids mirrored text on interchangeable SSD1306/SSD1315 modules. Both device-local preferences apply immediately, survive power cycling, and are preserved when presets or factory templates are loaded.

15 External SYNC and RST

The engine-side synchronization model is implemented, including pulse filtering, PPQN interpretation, loss policy and external reset semantics. The final comparator-to-timer capture routing and hardware-in-the-loop sign-off are still prerelease work.



SYNC setting	Options / meaning
SOURCE	INTERNAL · EXTERNAL · AUTO
PPQN	1 · 2 · 4 · 24 pulses per quarter note
EDGE	Rising or falling
LOSS	STOP · FREE (continue at last known tempo) · INTERNAL (fall back to internal BPM)
RESET INPUT	TRIGGER · GATE
FILTER	0...5000 µs in 250 µs steps
TIMEOUT	200...5000 ms in 100 ms steps

Source behavior

INTERNAL makes CLOCK the timing master. EXTERNAL follows accepted SYNC pulses. AUTO uses external timing while a usable external source is locked and internal timing otherwise.

RST behavior

In TRIGGER mode, the accepted reset edge resets global phase. In GATE mode, the external reset level controls the reset gate state. Channels whose local RESET policy is FREE are intentionally not forced back to the global cycle position.

PRERELEASE HARDWARE LIMITATION

Do not claim production-grade external sync from software tests alone. The final LM393/comparator input path, STM32 capture routing, thresholds and timing distribution must be measured on the real module. Until that HIL work is complete, INTERNAL is the reference operating path.

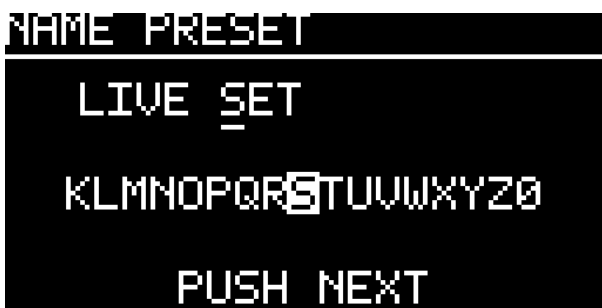
16 Presets, autosave and templates

CURRENT — automatic recovery

CLOCK continuously stages the complete working state. A coalescing delay avoids writing every small edit immediately. Physical STM32 Flash commits are deferred while transport is PLAYING and are flushed after PAUSE or STOP, so Flash erase/program work cannot stall the live gate path. On the next boot, a valid CURRENT state is restored but transport is still forced to STOP.

Eight named user presets

LOAD PRESET and SAVE PRESET expose eight durable user slots. Names are up to 16 characters. Saving over an occupied slot requires an explicit OVERWRITE PRESET confirmation with NO selected by default.



Factory templates

Template	Starting configuration
ALL MASTER	Factory One Clock state
CLOCK TREE	Independent: $\times 1$, $\div 2$, $\div 4$, $\div 8$, $\times 2$, $\times 4$, $\times 8$, $\times 16$
DIVIDERS	Independent: $\times 1$, $\div 2$, $\div 3$, $\div 4$, $\div 6$, $\div 8$, $\div 12$, $\div 16$
POLYRHYTHM	Ratios 1:1, 2:3, 3:2, 3:4, 4:3, 4:5, 5:4, 5:8

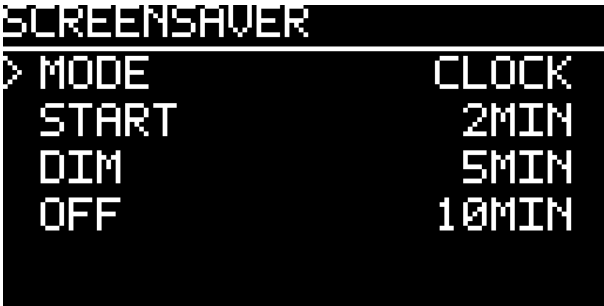
Template	Starting configuration
EUCLID KIT	Eight 16-step Euclid channels with differentiated hits/rotation
HYBRID	CH1–3 Clock, CH4–6 Euclid, CH7–8 Sequencer; CH2 $\div 2$, CH3 3:2

PRESET $\neq$ TEMPLATE

A named preset is your durable snapshot. A template is a factory starting point that replaces the current working configuration. Loading a template does not create a named preset.

17 Screensaver and display protection

Display protection advances only while transport is STOP. Any front-panel activity wakes the display immediately. PLAY and PAUSE do not accumulate the screensaver timers.



Factory timings

Stage	Factory value
Screensaver start	2 minutes
Dim	5 minutes
OLED power-off	10 minutes

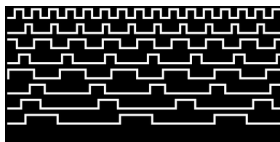
CLOCK enforces $START \leq DIM \leq OFF$. Choosing screensaver OFF disables the animation but does not disable the later DIM/OFF protection.

Available animations

Mode	Character
OFF	No animation; dim/off protection remains
CLOCK	Factory default; scrolling clock traces
PLUG	Damped plucked-string motion
HEARTBEAT	Pulsating heart
ACID	Bouncing/rotating acid-style smiley
SPECTRUM	Synthetic 16-band display; it does not

Mode	Character
	sample audio
FIELD	Slow monochrome contour field
BLOX	Procedural triangular bodies fall and stack
MATRIX	Original monochrome digital rain
CUBE COVER	8×8 cube tiles cover the OLED in changing traversal orders
FRACTAL	Curated Barnsley-fern crops
ORBIT	Sparse orbital animation

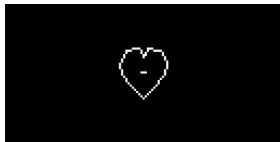
Screensaver gallery



CLOCK - scrolling clock traces



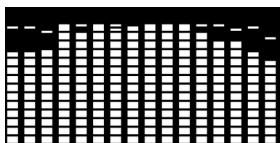
PLUG - damped plucked string



HEARTBEAT - pulsing trace



ACID - bouncing smiley



SPECTRUM - synthetic 16-band display



FIELD - moving contour field



BLOX - falling procedural bodies



MATRIX - monochrome digital rain



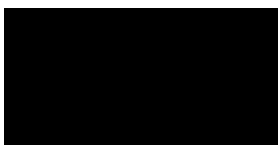
CUBE COVER - tiled cube
sweep



FRACTAL - curated fern
crop



ORBIT - sparse orbital
motion



OLED OFF - final protection
stage

18 Gate outputs and LEDs

Gate length

Gate length is selectable as 1, 2, 5, 10, 20, 50 or 100 ms; the factory value is 10 ms. The scheduler caps a pulse against the effective event interval so an excessively long setting cannot consume the next rising edge.

Voltage level

The current output-stage target is 0 V LOW and nominal +5 V HIGH. +10 V is not a project requirement; changing to that level would require a different output stage rather than the planned 74HCT244 path.

Activity LEDs

Each LED mirrors the corresponding logical channel signal. It is not a cable-presence indicator. Normal boot does not animate the output LEDs.

HARDWARE SIGN-OFF

Voltage levels, boot behavior at the jacks and timing jitter remain physical HIL measurements. Host tests verify logic, not analog thresholds or actual edge placement at a Eurorack socket.

19 Hidden boot features

Hold the encoder push continuously from power-up until the boot screen completes. CLOCK then starts the compile-time selected Easter egg before the normal real-time scheduler. The shipped/default selection is BEATKNECHT.

Build selection	Feature	Controls / behavior
1	Pixel Raid	Encoder moves; TAP fires; ranked arcade game
2	Formula 1	Encoder steers; ranked arcade game
3	Breakout	Encoder moves paddle; TAP launches; ranked arcade game
4	Egg Journey	Encoder moves; TAP jumps; ranked arcade game
5 (default)	BEATKNECHT	PLAY/PAUSE controls transport; STOP resets to step 1; TAP changes style; encoder changes BPM; long encoder push opens EXIT GAME?; drives OUT 1–8 intentionally

Easter-egg gallery



Pixel Raid - move and TAP to fire



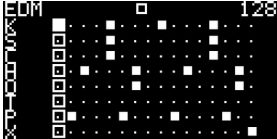
Formula 1 - steer through the course



Breakout - paddle, ball and modifiers



Egg Journey - jump through incoming hazards



BEATKNECHT - default eight-gate rhythm egg



Top 100 - ranked arcade leaderboards

Arcade flow

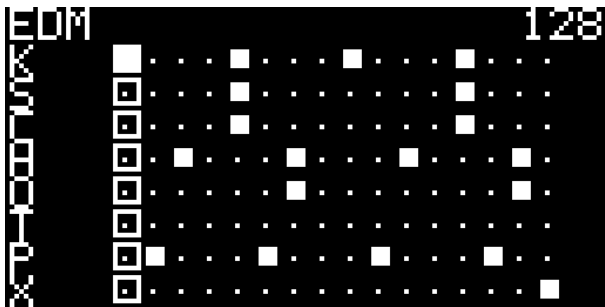
Pixel Raid, Formula 1, Breakout and Egg Journey share a game-specific intro, score/high-score tracking, three-letter initials entry when a score qualifies, and a scrollable Top 100. Their external gate-output stage stays disabled.



Exit

A long encoder hold is the exit gesture for the ranked arcade games. BEATKNECHT uses a guarded exit instead: a long encoder push opens EXIT GAME? with NO selected by default. Opening the prompt immediately silences all gates. NO returns to the exact previous transport state; if the rhythm was playing, playback resumes from the preserved phase, and if it was paused it remains paused. YES stops the

rhythm, resets the pattern, forces every gate LOW, disables the output stage, and returns to the normal clock firmware in STOP. The long press that opened the dialog cannot confirm it until the encoder has first been released.



20 Practical recipes

Eight exact copies of one master clock

- 1. Select One Clock.
- 2. Set RATE to $\times 1$ and HUMANIZE to OFF.
- 3. Choose the gate length.
- 4. Press PLAY.

A conventional clock tree

- 1. Switch to Independent.
- 2. Load the CLOCK TREE template or set channel rates manually.
- 3. Use RESET = GLOBAL on channels that must return to the same downbeat.

A Euclidean percussion line

- 1. Choose Euclid.
- 2. Set STEPS = 16.
- 3. Choose HITS.
- 4. Use ROTATE for placement.
- 5. Use PROBABILITY for controlled omissions.

A 64-step gate phrase

- 1. Choose Sequencer.
- 2. Set LENGTH = 64.
- 3. Open EDITOR.
- 4. Use PLAY/TAP to move forward/back through the four pages.
- 5. Toggle gates with encoder presses.
- 6. BACK returns to Sequencer settings.

21 Troubleshooting

Symptom	Check first
No gates after power-up	Normal until PLAY is pressed. Then check topology, OFF/MUTE, PROBABILITY and gate routing.
All eight outputs are identical	Factory topology is One Clock. Switch to a channel function to enter Independent operation.
Tempo stops at a value	Check GENERAL SETTINGS → CLOCK → MIN BPM / MAX BPM.
Tap Tempo changed clock source	Intentional: a valid Tap Tempo result selects INTERNAL.
RESET did not restart one channel	That channel may use RESET = FREE.
Screensaver does not start	Timers advance only in STOP, not PLAY or PAUSE.
Preset save asks for confirmation	The slot is occupied; NO is the safe default.
External lock is unavailable or unstable	External SYNC/RST hardware capture is prerelease and requires HIL sign-off; use INTERNAL as the reference path.

A USEFUL DIAGNOSTIC ORDER

First establish topology → transport → selected channel/mode → MUTE/OFF/PROBABILITY → rate → clock source. This avoids chasing timing settings when the problem is simply that no event is eligible to reach the output.

22 Firmware installation and updates

CLOCK uses an STM32F401 Black Pill. There are two supported programming paths: ST-LINK/SWD is the most robust choice for a first installation or recovery, while the built-in USB DFU bootloader is the convenient path for routine updates. The procedures below deliberately preserve the Flash areas used for settings, presets and high scores.

USB POWER SAFETY

Never connect the Black Pill USB port while CLOCK is also powered from the Eurorack bus. Before any USB update or USB-only test, switch the rack off and preferably unplug CLOCK's Eurorack ribbon cable. Leave the rack cable disconnected until the USB cable has been removed.

Which method should I use?

Use ST-LINK for a blank Black Pill, first installation, recovery from a failed flash, or debugging. Use USB DFU for normal firmware updates when the module already works and you simply want to load a newer build.

Method A — ST-LINK / SWD

ST-LINK talks directly to the STM32 debug interface and does not depend on the installed firmware. This makes it the preferred recovery path. The example below uses STM32CubeProgrammer to write the address-aware ELF generated by PlatformIO.

- 1. Switch the Eurorack power off. Use exactly one target power source. If you power the Black Pill from its USB connector while programming, unplug the Eurorack ribbon cable first.
- 2. Connect ST-LINK GND to GND, SWDIO to SWDIO, and SWCLK to SWCLK. Connect the probe

target-voltage reference (VTref / VDD_TARGET) to the Black Pill 3.3 V rail. NRST is optional but recommended for reliable recovery. On an official ST-LINK, VTref is a target-voltage reference; do not assume that it powers the module.

- 3. Install PlatformIO and open the CLOCK source tree. Build the required firmware, for example:
`pio run -e release_default`
- 4. The resulting address-aware image is `.pio/build/release_default/firmware.elf`. In STM32CubeProgrammer choose ST-LINK and SWD, connect to the target, open that ELF file, then use Download/Program with verification enabled.
- 5. Reset or power-cycle the Black Pill after programming. For a normal update, do not use Full chip erase: a full erase also removes settings, presets and high scores.

The `release_default` environment builds the normal BEATKNECHT image. The alternative build environments are `release_pixel_raid`, `release_formula_1`, `release_breakout` and `release_egg_journey`.

Method B — USB DFU with PlatformIO

The STM32F401 contains an ST system-memory USB DFU bootloader, so routine updates do not require an external programmer. CLOCK's PlatformIO configuration uses a project-specific upload helper that programs only the firmware-owned Flash regions and leaves the A/B persistence sectors untouched.

- 1. Switch the Eurorack power off and unplug CLOCK's Eurorack ribbon cable. This is mandatory before connecting USB.
- 2. Use a data-capable USB cable and connect the Black Pill to the computer.

- 3. Enter DFU mode: hold BOOT0, press and release NRST/RESET while BOOT0 remains held, wait roughly half a second, then release BOOT0. Alternatively, with the board unpowered, hold BOOT0 while connecting USB and release it after roughly half a second.
- 4. Open a terminal in the CLOCK source directory and run: `pio run -e release_default -t upload`
- 5. Wait for PlatformIO to report a successful upload. The uploader writes the application region at 0x0800C000 first, writes sector 0/vector table at 0x08000000 last, leaves persistence sectors 1 and 2 untouched, then asks the ROM bootloader to leave DFU mode from the vector-table address. If the board does not restart automatically, make sure BOOT0 is released and press RESET once.
- 6. Leave the Eurorack ribbon cable disconnected for the first functional check. CLOCK can boot from the Black Pill USB supply, so you can verify the display, encoder, buttons, menus and general UI before returning the module to the rack.
- 7. When finished, disconnect USB first. Then reconnect the Eurorack ribbon cable with the red stripe at -12 V and only then power the rack again.

DO NOT FLASH A FLAT FIRMWARE.BIN

CLOCK reserves Flash sectors 1 and 2 for power-loss-tolerant A/B persistence. A conventional contiguous firmware.bin can span that reserved gap and destroy saved state. Use the project PlatformIO upload command, the release DfuSe image with a DfuSe-aware tool, or the address-aware firmware.elf over ST-LINK.

PlatformIO installation

For most DIY builders, PlatformIO IDE for Visual Studio Code is the easiest installation because PlatformIO Core is

included. If you prefer the command line, install PlatformIO Core separately. Official installation documentation: <https://docs.platformio.org/en/stable/integration/ide/vscode.html> and

<https://docs.platformio.org/en/stable/core/installation/>

On Windows, an STM32 device in ROM DFU mode should appear as an STM32 bootloader/DFU device (ST VID:PID 0483:DF11). If the driver is missing, install the current STM32CubeProgrammer package and its DFU driver. On Linux, install PlatformIO's current udev rules as described in the PlatformIO documentation.

If an update is not detected

- 1. Confirm that the USB cable carries data; charge-only cables are a common cause of apparently dead DFU connections.
- 2. Repeat the BOOT0/RESET sequence and check that the host sees VID:PID 0483:DF11 before running the upload command.
- 3. If ST-LINK cannot connect, first check target power, common GND, VTref, SWDIO and SWCLK; add NRST when recovering a target that behaves unexpectedly.
- 4. If settings or presets disappear after an update, check whether a Full chip erase or a flat contiguous binary was used. The normal CLOCK update paths are designed to preserve persistence.

Timing architecture

The current prerelease uses a 20 kHz TIM3 scheduler (50 μ s service quantum) and no RTOS. Conditioned SYNC/RST edges are captured through GPIO EXTI and consumed by the scheduler. The physical PEC11L encoder uses PA0/PA1 in STM32 TIM2 encoder mode, so quadrature transitions are counted in hardware rather than through encoder EXTI. Display and foreground work remain subordinate to the musical timing path.

Display transports

SPI is the preferred/default reference display path and performs fast dirty-page transfers. I²C remains supported, but display work is deliberately bounded and deferred: present() only publishes the latest framebuffer, at most one I²C transaction is serviced per foreground pass, data packets contain at most 24 framebuffer bytes, dirty pages are used, stale UI frames may be dropped, and NACKs are retried. Display refresh is never allowed to decide gate timing.

REQUIRED HIL A/B TEST

Before 1.0 hardware sign-off, SPI and I²C must be compared on the real STM32 under maximum display activity. A lower I²C frame rate is acceptable; additional gate jitter, lost SYNC/RST events or encoder detents are not. The automated encoder regressions exercise first-detent, direction-reversal, wraparound and corrupted-transition recovery, but the remaining first-detent observation reported on physical hardware remains a HIL item until confirmed on the module.

Prototype specifications

Parameter	Current documented value
Format	Eurorack, 3U
Width	10 HP
MCU	STM32F401CCU6 Blackpill
Display	128×64 SSD1306/SSD1315; SPI reference, I ² C supported
Outputs	8 gate/clock outputs; target 0/+5 V
Inputs	SYNC and RST, final hardware capture/routing prerelease
Controls	Push encoder, PLAY, TAP, STOP/BACK, 8 activity LEDs
Scheduler	20 kHz hard timing domain
Persistence	CURRENT auto-save + 8 named presets, schema/CRC protected
Firmware	0.19.0-beta.14 (prototype)

Verification boundary

The source release exposes 390 named Native tests plus sanitizer, simulator, architecture, dependency and documentation gates. Those checks validate software behavior. They do not prove comparator thresholds, electrical output levels, physical gate jitter, mechanical encoder behavior, startup pulses at the jack or final SPI/I²C signal integrity. Those remain HIL measurements.

COLOPHON

CLOCK — User Manual. Typeset in Ubuntu.

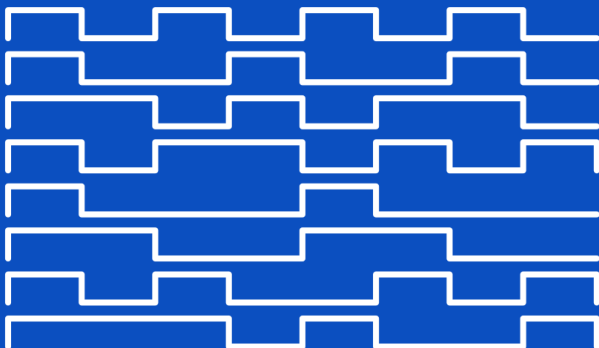


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Design, firmware and documentation: Axel Napolitano.

Prototype documentation — firmware 0.19.0-beta.14, baseline 16 September 2026.

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CLOCK

Eight-channel clock & gate generator

Design & firmware

Axel Napolitano

Width · format

10 HP · Eurorack 3U

Firmware

0.19.0-beta.14 (beta)

Firmware licence

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